

COTTONWOOD PASS, CO, USA

WMO: 726391

Lat: 38.782N

Lon: 106.218W

Elev: 2995

StdP: 70.15

Time Zone: -7.00 (NAM)

Period: 05-19

WBAN: 00385

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-16.2	-13.0	-38.0	0.1	3.5	-33.0	0.2	2.7	17.8	-0.3	15.6	-1.2	2.5	270	0.627

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	27.0	8.7	25.1	8.4	23.0	8.1	12.5	18.7	11.7	18.4	11.0	17.9	2.4	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.9	11.8	14.8	9.2	10.5	14.1	7.8	9.5	14.1	45.4	18.7	42.8	18.1	40.9	18.0	15.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.9	11.5	9.6		-18.3	29.9	1.6	0.9	-19.5	30.6	-20.4	31.1	-21.3	31.6	-22.4	32.3
				-19.8	14.4	2.9	1.0	-22.0	15.1	-23.7	15.7	-25.3	16.2	-27.5	17.0
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.9	-2.3	-2.3	1.5	4.2	8.9	16.2	18.4	16.9	13.7	7.2	2.6	-2.5		
	DBStd	8.71	4.89	5.05	4.82	4.29	4.53	3.89	2.31	2.33	3.60	4.85	5.17	5.28		
	HDD10.0	1983	383	346	265	177	76	6	0	0	12	109	223	388		
	HDD18.3	4228	641	579	523	423	292	81	28	54	143	345	472	647		
	CDD10.0	858	0	0	1	3	42	192	259	215	122	23	1	0		
	CDD18.3	61	0	0	0	0	0	18	29	11	3	0	0	0		
	CDH23.3	369	0	0	0	0	0	128	169	56	16	0	0	0		
	CDH26.7	36	0	0	0	0	0	15	19	2	1	0	0	0		
(14) Wind	WSAvg	3.5	3.9	4.6	4.0	3.9	3.1	3.0	2.4	2.4	2.8	3.6	4.0	4.1		
(15) Precipitation	PrecAvg	287	11	13	20	30	27	20	43	53	27	22	14	11		
	PrecMax	356	24	31	47	68	79	37	86	99	51	59	34	58		
	PrecMin	194	2	3	4	5	2	7	7	17	9	6	4	4		
	PrecStd	48	6	9	11	20	20	9	22	18	12	13	7	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.1	10.3	15.2	17.4	22.4	28.0	28.1	27.0	25.9	20.0	15.0	9.9		
		MCWB	0.6	0.2	2.4	3.9	6.0	8.0	9.3	8.8	7.8	5.5	2.6	-0.5		
	2%	DB	7.3	7.7	12.3	15.0	20.1	26.4	27.1	25.0	22.8	17.5	12.5	7.3		
		MCWB	-0.9	-0.7	1.3	2.6	5.3	7.8	9.3	8.6	7.1	4.6	1.9	-1.5		
	5%	DB	5.8	6.1	10.2	12.6	17.8	24.9	25.2	22.9	21.4	16.0	11.1	5.9		
		MCWB	-1.7	-1.6	0.4	1.9	4.6	7.4	9.3	8.6	7.0	4.1	1.3	-1.8		
	10%	DB	3.0	4.0	7.8	11.2	16.3	22.7	23.0	22.2	19.9	13.7	9.0	3.8		
		MCWB	-2.9	-2.4	-0.5	1.5	4.2	6.9	9.1	8.7	6.8	3.6	0.4	-2.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.8	1.6	3.4	5.3	8.0	11.6	14.3	13.3	11.5	8.8	4.3	1.3		
		MCDB	6.8	7.1	12.1	12.4	15.3	18.8	20.4	18.9	17.4	13.5	10.5	5.5		
	2%	WB	0.2	0.2	1.9	4.2	6.9	10.3	12.9	12.2	10.3	7.0	2.8	0.1		
		MCDB	5.1	5.4	9.5	11.5	15.2	19.0	18.9	18.3	16.2	13.0	10.0	4.4		
	5%	WB	-0.9	-0.7	1.1	3.0	6.1	9.5	12.1	11.4	9.3	5.7	1.9	-0.9		
		MCDB	4.1	4.4	8.3	10.4	14.6	18.7	18.7	17.8	16.2	12.1	8.7	3.9		
	10%	WB	-2.0	-1.6	0.2	2.1	5.3	8.7	11.4	10.7	8.5	4.7	0.9	-1.8		
		MCDB	2.7	3.5	6.9	8.9	13.1	18.8	18.5	17.3	15.6	11.6	8.1	3.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	7.2	8.6	9.6	10.4	11.9	11.8	10.8	10.7	9.2	7.9	6.6		
		MCDBR	7.7	8.0	10.0	11.2	11.8	13.3	13.1	11.8	11.8	10.5	8.9	7.0		
	5% WB	MCWBR	4.5	4.3	4.6	4.9	4.6	4.8	4.1	4.1	4.2	4.5	4.2	3.9		
		MCWBR	6.9	7.0	9.1	10.5	11.4	12.7	11.4	10.4	10.0	9.5	8.3	6.4		
(40) Clear-Sky Solar Irradiance	taub	taub	0.206	0.214	0.237	0.258	0.267	0.270	0.307	0.294	0.256	0.239	0.217	0.208		
		taud	2.497	2.461	2.461	2.514	2.597	2.618	2.537	2.559	2.703	2.685	2.635	2.530		
	Ebn at Noon	Ebn at Noon	1014	1041	1037	1023	1012	1004	963	973	1005	1001	996	990		
		Edh at Noon	74	89	100	102	96	95	102	97	78	71	65	67		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2.76	3.71	5.07	5.93	6.43	7.29	6.40	5.78	5.26	4.03	2.94	2.37		
	RadStd	RadStd	0.14	0.21	0.38	0.37	0.49	0.44	0.35	0.28	0.33	0.28	0.19	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+73	+20

Nomenclature: See separate page